

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>
She has 5 vacuum cleaners left
She sold half of what was left at the orange house so before selling to orange house she had $5 \times 2 = 10$ vacuum cleaners
She sold 2 more to the red house so before selling to red house she had $10 + 2 = 12$ vacuum cleaners
She sold a third at the green house so 12 is the two thirds that she had after selling at green house the number she sold at green house is $12 - \frac{2}{3}(12) = 4$
She started with $3(4) = 12$ vacuum cleaners
</think>
<answer>12</answer>

